

# PowerShell Automation & Reporting Scripts

Professional PowerShell scripts for Active Directory and Microsoft 365 administration, automation, and compliance reporting.

## Overview

This repository contains production-ready PowerShell scripts designed to streamline enterprise IT operations, reduce manual work, and provide security/compliance insights.

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## Scripts

### 1. Create-BulkADUsers.ps1

**Purpose:** Bulk create Active Directory user accounts from a CSV file

**Use Cases:** Employee onboarding, batch user provisioning, organizational restructuring

**Features:**

- Reads user data from CSV (FirstName, LastName, Department, Email, Manager)
- Auto-generates usernames following naming conventions
- Creates temporary passwords with forced change on first logon
- Validates target OU exists before processing
- Skips existing users gracefully
- Generates detailed CSV report with success/failure tracking
- Color-coded console output for easy monitoring

**Requirements:**

- Active Directory module (Windows Server)
- Domain Administrator privileges
- CSV file with columns: FirstName, LastName, Department, Email, Manager

**Usage:**

```
# Create users from CSV in specific OU
.\Create-BulkADUsers.ps1 -CSVPath "C:\users.csv" -TargetOU "OU=NewEmployees,OU=Us

# Use default OU (OU=Users,DC=contoso,DC=com)
.\Create-BulkADUsers.ps1 -CSVPath "C:\users.csv"
```

### Example CSV Format:

```
FirstName,LastName,Department,Email,Manager
John,Smith,Engineering,john.smith@contoso.com,Jane.Doe
Sarah,Johnson,Marketing,sarah.johnson@contoso.com,Mark.Wilson
```

### Output:

- Console log with real-time creation status
  - CSV report: BulkUserCreation\_Report\_[timestamp].csv
  - Each created user gets temporary password & must change on next logon
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## 2. Get-ADUserAudit.ps1

**Purpose:** Audit Active Directory for inactive users, stale passwords, and security risks

**Use Cases:** Compliance reporting, account hygiene, security reviews, identifying accounts to deactivate

### Features:

- Identifies inactive user accounts (configurable threshold, default 90 days)
- Flags stale passwords (configurable threshold, default 120 days)
- Calculates risk levels (High = inactive + stale password)
- Handles never-logged-in accounts
- Displays top 10 findings by risk category
- Exports comprehensive audit report to CSV
- Color-coded risk indicators

### Requirements:

- Active Directory module

- Domain Administrator or similar read privileges
- Run as Administrator

### Usage:

```
# Run audit with default thresholds (90 days inactive, 120 days stale password)
.\Get-ADUserAudit.ps1 -ExportToCSV

# Custom thresholds - flag inactive after 60 days, stale passwords after 90 days
.\Get-ADUserAudit.ps1 -InactiveDays 60 -StalePasswordDays 90 -ExportToCSV

# Audit specific OU only
.\Get-ADUserAudit.ps1 -TargetOU "OU=Engineering,DC=contoso,DC=com" -ExportToCSV
```

### Output:

- Console summary with high-level statistics
- Top 10 inactive accounts (days since login)
- Top 10 stale password accounts (days since password change)
- CSV report: ADUserAudit\_[timestamp].csv (if -ExportToCSV specified)

### Typical Findings Report Columns:

- Username, DisplayName, Email, Department
- LastLogonDate, DaysInactive
- PasswordLastSet, DaysSincePasswordChange
- NeverLoggedIn (boolean)
- RiskLevel (Low/Medium/High)

### Use Cases:

- Quarterly compliance reviews
- Identifying accounts to disable
- Password policy enforcement
- Detecting compromised accounts

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## 3. Get-M365MailboxAudit.ps1

**Purpose:** Audit Microsoft 365 users and licensing compliance

**Use Cases:** License optimization, user activation tracking, compliance reporting

**Features:**

- Retrieves all Microsoft 365 users and their assigned licenses
- Maps license SKUs to human-readable names (E3, E5, F1, etc.)
- Identifies unlicensed users
- Counts active licenses by type
- Optional filtering by license type
- Exports separate reports for licensed/unlicensed users
- Provides license optimization recommendations

**Requirements:**

- Microsoft.Graph PowerShell module ( `Install-Module Microsoft.Graph` )
- Microsoft 365 admin account with appropriate permissions
- Scopes: `Organization.Read.All`, `User.Read.All`

**Installation:**

```
# Install Microsoft.Graph module (one-time)
Install-Module Microsoft.Graph -Scope CurrentUser

# Grant required permissions
Connect-MgGraph -Scopes "Organization.Read.All", "User.Read.All"
```

**Usage:**

```
# Audit all users and licenses
.\Get-M365MailboxAudit.ps1 -ExportToCSV

# Filter by specific license type (e.g., only E5 licenses)
.\Get-M365MailboxAudit.ps1 -LicenseFilter "E5" -ExportToCSV

# Without export (console output only)
.\Get-M365MailboxAudit.ps1
```

**Output:**

- **Console summary:**
    - **Total users**
    - **Licensed users count**
    - **Unlicensed users count**
    - **License distribution by type**
  - **CSV reports (if -ExportToCSV):**
    - M365\_LicensedUsers\_[timestamp].csv
    - M365\_UnlicensedUsers\_[timestamp].csv (if applicable)
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## Common Scenarios

### Scenario 1: Onboarding 50 New Employees

```
# 1. Prepare CSV with employee data
# 2. Run bulk creation script
.\Create-BulkADUsers.ps1 -CSVPath "C:\new_employees_2024.csv"

# 3. Review report and share temporary passwords securely
# 4. Employees log in and change passwords
```

### Scenario 2: Quarterly Compliance Review

```
# 1. Audit AD for inactive accounts
.\Get-ADUserAudit.ps1 -InactiveDays 90 -ExportToCSV

# 2. Review findings
# 3. Disable or archive flagged accounts
# 4. Follow up with account owners
```

### Scenario 3: License Cost Optimization

```
# 1. Audit M365 licenses
.\Get-M365MailboxAudit.ps1 -ExportToCSV

# 2. Analyze unlicensed users - are they active?
# 3. Identify over-licensed users (E5 when E3 sufficient)
```

## Best Practices

### Safety & Governance

- **Always test on a small sample first** before running bulk operations
- **Keep audit reports** for compliance documentation
- **Schedule audits monthly** and track trends over time
- **Use -WhatIf parameter** when available (future versions)

### Performance

- For large environments (1000+ users), consider filtering by OU
- Schedule audits during non-business hours
- Archive old CSV reports to a compliance folder

### Security

- Store CSV input files securely (sensitive data like email domains)
  - Protect generated reports (contain passwords, user info)
  - Limit script execution to privileged admins
  - Review audit logs for script execution
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## Troubleshooting

### “Active Directory module not found”

- **Solution:** Run script as Administrator on a domain-joined server
- **Run:** `Import-Module ActiveDirectory` first

### “Access Denied” errors in bulk user creation

- **Solution:** Ensure you have Domain Admin or delegated OU permissions

- **Check target OU path exists:** `Get-ADOrganizationalUnit -Identity "OU=Users,DC=contoso,DC=com"`

## **"Cannot connect to Microsoft 365"**

- **Solution:** Install Microsoft.Graph module and authenticate first

```
Install-Module Microsoft.Graph
Connect-MgGraph -Scopes "Organization.Read.All", "User.Read.All"
```

## **CSV encoding issues**

- **Solution:** Save CSV as UTF-8 without BOM in Excel
  - **Use:** File → Save As → CSV (Comma delimited) → Tools → Web Options → Encoding: UTF-8
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## **Contributing**

These scripts are provided as-is for portfolio demonstration. Feedback and improvements welcome.

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## **License**

Personal use for IT administration and compliance reporting.

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PowerShell | Identity Management | Compliance & Reporting

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## **Version History**

- **v1.0** (2024) - Initial release
  - Create-BulkADUsers.ps1

- **Get-ADUserAudit.ps1**
- **Get-M365MailboxAudit.ps1**